

ML Assignment

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1 Introduction

This report contains details regarding the polynomial regression models for two phases of a geothermal power plant expansion project. The objective is to obtain accurate predictions on the provided test datasets using only polynomial regression. Phase 1 focuses on optimizing a multi-stage steam turbine system based on six operational parameters. Phase 2 involves mapping a high-potential geological survey block using three spatial coordinate offsets to predict a Thermal Anomaly Score. Performance is measured using Mean Squared Error (MSE) and the R^2 Score (Coefficient of Determination).

2 Methodology

To prevent data leakage and to ensure generalization, an 80/20 train-validation split was used. The data matrix was constructed using the `itertools.combinations_with_replacement` function to generate all possible polynomial interactions on the fly.

Two distinct modeling strategies were evaluated for both phases:

1. **Standard Polynomial Regression (SPR):** A brute-force approach of complete polynomial degrees using the Moore-Penrose pseudo-inverse (i.e the normal equation for each degree).
2. **L1 Regularization (Lasso):** A max degree was chosen and L1 regularization was implemented, which essentially acts as a feature selector. Inbuilt methods were used for this.

3 Phase 1: Power Plant Steam Turbine Optimization (var1)

The Net Power Score is governed by six unique operational parameters. The assignment bounds the theoretical polynomial model up to a moderate degree of 10.

3.1 Standard Polynomial Regression

Sequential evaluation of complete polynomial matrices revealed that a Degree 4 model had the optimal output (Table 1), achieving a Validation MSE of 0.7891 and an R^2 of 0.9209. Moving onto Degree 5 introduced immediate overfitting, raising the MSE to 1.2356. And at Degree 6, the model fell victim to the curse of dimensionality (924 features for 800 training samples), causing the validation R^2 to collapse to -21.98.

Degree	Validation R^2	Validation MSE	Total Features
Degree 2	0.6798	3.1945	28
Degree 3	0.8762	1.2345	84
Degree 4	0.9209	0.7891	210
Degree 5	0.8761	1.2356	462
Degree 6	-21.9892	229.4189	924

Table 1: SPR Performance across Degrees

3.2 L1 Regularization and Feature Sparsity

To extract higher-order features without the accompanying noise and overfitting, an L1 regularized model was trained on the full Degree 10 expansion (8,008 features).

The optimal L1 penalty (λ) was found through an iterative, narrowing hyperparameter grid search:

- An initial logarithmic sweep across [0.0001, 0.001, 0.01, 0.1, 1, 10] identified 0.01 as the most promising magnitude.
- The next sweep across [0.005, 0.01, 0.03, 0.05] identified 0.005 as the best value.
- A final search across [0.002, 0.003, 0.004, 0.0045, 0.005, 0.0055, 0.006, 0.007, 0.008] confirmed $\lambda = 0.005$ as the optimal value.

This optimal Lasso model achieved a better Validation MSE of 0.3684 and an R^2 of 0.9631, keeping exactly 125 features from the 8,008-column matrix. The breakdown of active features retained per degree is detailed below.

Polynomial Degree	Number of Active Features
Degree 10	17
Degree 9	20
Degree 8	5
Degree 7	12
Degree 6	5
Degree 5	22
Degree 4	17
Degree 3	13
Degree 2	10
Degree 1	3
Degree 0 (Bias)	1

Table 2: Lasso Feature Retention by Degree

3.3 Rationale and Final Selection

By prioritizing feature-level sparsity over rigid degree-level cutoffs, the Lasso model significantly outperformed the standard polynomial regression model.

Model Strategy	Validation R^2	Validation MSE	Active Features
Standard PR (Degree 4)	0.9209	0.7891	210
Lasso ($\lambda = 0.005$)	0.9631	0.3684	125

Table 3: Phase 1 Model Comparison

Interestingly, the L1 regularizer retained 17 highly specific terms from the 10th degree. This demonstrates that while a *complete* degree 10 matrix heavily overfits (as seen in the SPR collapse), some high-order interaction terms actually carry useful information regarding the prediction of the turbine’s unique Net Power Score. The Lasso model effectively extracted these 125 optimal curves while zeroing out 7,883 noise columns, making it the optimal choice for the final Phase 1 predictions.

4 Phase 2: Subterranean Thermal Reservoir Mapping (var2)

This phase maps geological structures using 3D spatial coordinate offsets (x_1, x_2, x_3) . Because of the complex geological environment, the underlying heat map resembles a high-degree polynomial up to degree 20.

4.1 Standard Polynomial Regression Evaluation

The SPR evaluation peaked at Degree 8 (Table 4), achieving a Validation MSE of 0.2702 and an R^2 of 0.9935. Pushing to Degree 9 triggered overfitting, slightly elevating the MSE to 0.2896, and by Degree 14, the variance got out of hand (overfitting), generating an R^2 of -3.15.

Degree	Validation R^2	Validation MSE	Total Features
Degree 6	0.9875	0.5200	84
Degree 7	0.9921	0.3275	120
Degree 8	0.9935	0.2702	165
Degree 9	0.9930	0.2896	220
Degree 10	0.9903	0.4013	286

Table 4: Performance across Degrees

4.2 L1 Regularization and Feature Sparsity

A full Degree 20 expansion was constructed, yielding a 1,771-feature design matrix. And Lasso was used.

The optimal L1 penalty (λ) was again identified through a targeted search:

- An initial logarithmic sweep across [0.0001, 0.001, 0.01, 0.1, 1, 10] indicated the optimal value was between 0.0001 and 0.001.
- The next sweep [0.0002, 0.0003, 0.0005, 0.0007, 0.001, 0.0015] gave the optimal value as 0.0003.
- The final sweep from 0.00022 to 0.0004 gave the optimal value as $\lambda = 0.00036$.

The optimal Lasso model achieved a Validation MSE of 0.2517 and an R^2 of 0.9940, retaining 237 essential features. The no of features present for each degree is as follows.

Polynomial Degree	Number of Active Features
Degree 20	22
Degree 19	26
Degree 18	10
Degree 17	13
Degree 16	11
Degree 15	14
Degree 14	11
Degree 13	12
Degree 12	12
Degree 11	11
Degree 10	13
Degree 9	16
Degree 8	12
Degree 7	14
Degree 6	10
Degree 5	10
Degree 4	7
Degree 3	6
Degree 2	3
Degree 1	3
Degree 0 (Bias)	1

Table 5: Lasso Feature Retention by Degree

4.3 Rationale and Final Selection

The L1 regularizer only marginally improved upon the already highly accurate Degree 8 SPR model.

Model Strategy	Validation R^2	Validation MSE	Active Features
Standard PR (Degree 8)	0.9935	0.2702	165
Lasso ($\lambda = 0.00036$)	0.9940	0.2517	237

Table 6: Phase 2 Model Comparison

5 Conclusion

The usage of L1 regularization on the theoretical maximum-degree data matrix expansions gave better results than standard SPR that had degree level cutoffs for both the phases. The final models achieved R^2 scores of 0.9631 and 0.9940 for Phase 1 and Phase 2, respectively. Both predictions have been generated and formatted as `<ROLLNO>_pred_var<PRBID>.csv`.

The Link to the Github Repository: [GitHub Repository](#)